Bharat AI Platform

User Manual

Version 1.0

Centre for Development of Advanced Computing (C-DAC)
Chennai

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1. Introduction

Artificial Intelligence (AI) is emerging as a critical technology for enabling innovation, digital transformation, intelligent decision-making, and the delivery of advanced technology-driven services. As AI adoption expands across sectors, the availability of secure, scalable, and sovereign AI infrastructure has become increasingly important for building indigenous technological capabilities.

The **Bharat AI Platform** is developed in the spirit of **Atmanirbhar Bharat**, with the objective of strengthening India's capabilities in Artificial Intelligence through sovereign infrastructure and indigenous AI services.

The platform provides an environment for hosting and delivering advanced AI models and services using dedicated GPU-based computing infrastructure. The platform is designed to support the deployment and scaling of large-scale AI workloads while maintaining control over the underlying computing infrastructure.

The Bharat AI Platform has been deployed by the **Centre for Development of Advanced Computing (C-DAC), Chennai**, as part of the broader effort to strengthen India's indigenous AI ecosystem. C-DAC is a premier R&D organization under the Ministry of Electronics and Information Technology (MeitY), with established expertise in high-performance computing, Artificial Intelligence, multilingual computing, and related technologies.

The platform represents an important step towards providing access to advanced AI capabilities through infrastructure operated within India's technological ecosystem. It brings together AI models, GPU computing resources, and platform services to support AI-based applications and use cases.

The Bharat AI Platform is intended to serve as a foundation for users and organizations seeking to leverage advanced AI capabilities while contributing to the development of a strong, self-reliant, and sovereign AI ecosystem in India.

2. About Bharat AI

2.1. Overview

Bharat AI is a sovereign AI platform developed to provide access to advanced Artificial Intelligence capabilities through infrastructure established within India's AI ecosystem and is aligned with the broader national objective of developing an indigenous and sovereign AI ecosystem. The National Portal of India describes the Bharat AI Platform in the context of India's rapid development of a sovereign, indigenous AI ecosystem designed to address India's multilingual and sectoral requirements. The platform is designed to bring advanced AI capabilities closer to users through a unified environment for accessing AI services and models.

2.2. Atmanirbhar Bharat and Sovereign AI

The Bharat AI Platform is developed in the spirit of **Atmanirbhar Bharat**, emphasizing technological self-reliance and the development of critical digital capabilities within the country.

A sovereign AI ecosystem enables greater control over the infrastructure, computing resources, AI models, and associated technology stack used to deliver AI capabilities. The platform contributes to this objective by providing AI capabilities on dedicated computing infrastructure and by enabling large-scale AI workloads to be executed within India's technological infrastructure. This approach is particularly relevant for applications where control over computing infrastructure, data processing, security, scalability, and operational management is important.

2.3. Large Language Models & Gen AI

The platform is intended to provide access to advanced AI models, including Large Language Models (LLMs), through its AI infrastructure.

Large Language Models are AI models capable of processing and generating human-language content. Depending on the model and service, such capabilities may include:

- Natural language understanding
- Text generation
- Question answering
- Summarization
- Information extraction
- Translation
- Conversational interaction
- Content generation
- Other AI-assisted tasks

The actual models and AI services available through the Bharat AI Platform may evolve over time as new models and capabilities are introduced. Accordingly, users should refer to the platform interface and the latest official platform documentation for the currently available models and services.

3. Guidelines for Effective Use of AI Services

AI Model / Service Selection

The Bharat AI Platform provides multiple AI services designed for different user requirements. Users can select the appropriate service based on the nature of the task they want to perform.

The available services shown on the platform are:

1. General Purpose
2. PPT Generation
3. Summarization
4. Document Generation
5. Multilingual Reasoning
6. Cyber Security

The following sections describe the purpose, selection procedure, and supported input types for each service.

1. General Purpose

1.1. Purpose and Use

The **General Purpose** service is intended for general-purpose AI interaction and can be used for a wide range of question-answering, reasoning, content-generation, technical, and conversational tasks.

The General Purpose service can be used for:

- General question answering
- General knowledge queries
- Technical questions
- Non-technical questions
- User-intent understanding
- Complex queries and problem solving
- Multi-turn conversations
- Ambiguous queries
- Instruction-based tasks
- Mathematical and logical reasoning
- Content generation

- Factual question answering
- Meeting-notes analysis
- Government-order summarization and question answering
- Summarization
- Multilingual queries

The **General Purpose** service is designed to provide detailed responses, follow user instructions, maintain context across multi-turn interactions, and support queries in multiple languages.

1.2. When to Select General Purpose

General Purpose should be selected for tasks that require general AI assistance, including question answering, reasoning, content generation, analysis, and conversational interactions, rather than tasks specifically intended for presentation generation, document generation, summarization, multilingual reasoning, or cybersecurity.

Examples include:

"Explain the difference between a process and a thread."
"Explain REST API in five bullet points."
"How do I learn Python for data analysis?"
"Design a web application using React, Python, and PostgreSQL."
"What is the time complexity of binary search?"
"Help me draft an email to a software development team."

1.3. Selecting General Purpose for General AI Assistance

Procedure

1. Log in to the Bharat AI Platform - https://bharatai.gov.in
2. Open the **AI Service Selection** menu.
3. Select **General Purpose**.
4. Verify that **General Purpose** is displayed as the selected service.
5. Enter the question, instruction, or task in the input area.

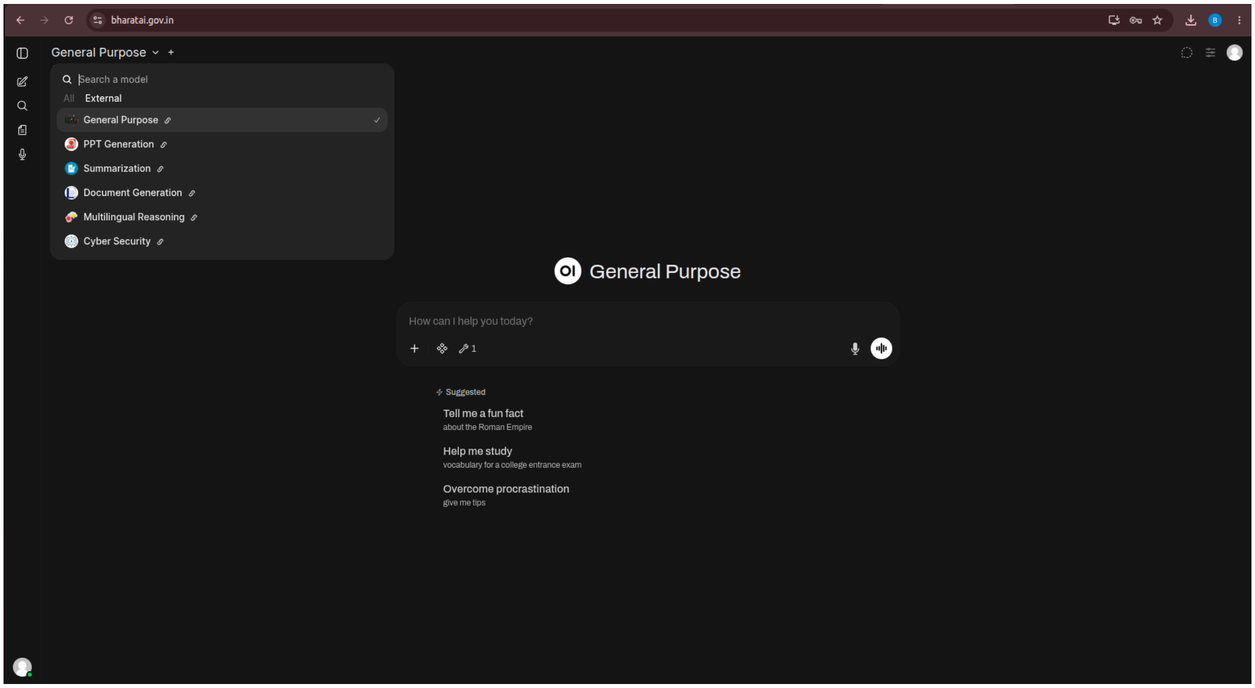


Figure: General Purpose Service Selection

1.4. Input Types

The **General Purpose** service supports various types of user inputs for performing different AI-assisted tasks.

A. Text Input

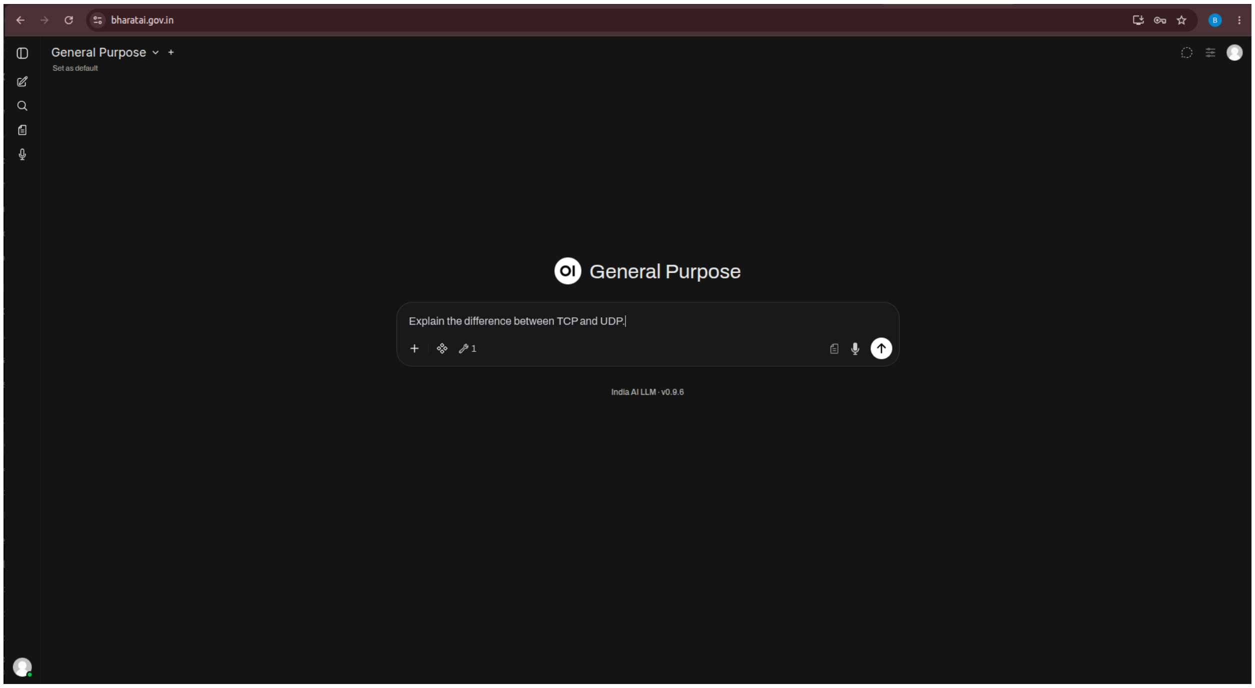
Text is the primary input method. Users can enter questions, instructions, or other textual information to interact with the service.

Typical text inputs include:

- General knowledge questions
- Technical queries
- Instructions and prompts
- Mathematical and logical problems
- Coding-related queries
- Content-generation requests
- Requests for explanations
- Follow-up questions in an ongoing conversation

Example:

> "Explain the difference between TCP and UDP."



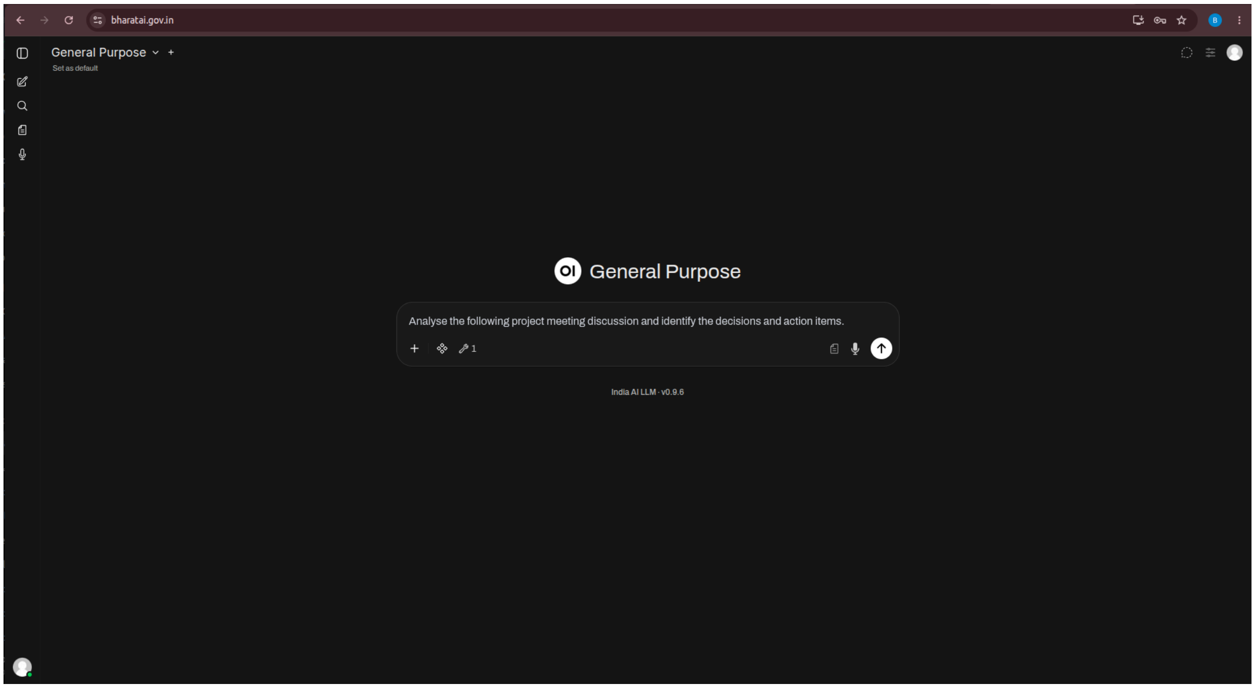
B. Long Text / Structured Text

Users can provide **long-form or structured textual content** for analysis and processing. The content can be used for tasks such as:

- Content analysis
- Summarization
- Question answering
- Information extraction
- Reasoning and interpretation

Example:

> "Analyse the following project meeting discussion and identify the decisions and action items."



C. PDF / Document Input

Users can upload **supported PDF documents** for content analysis and processing. The uploaded document can be used to extract relevant information, answer questions, explain content, analyse information, or generate responses based on the document.

Example:

> Upload a PDF document and ask: "Summarize the key points discussed in this document."

Common document-based tasks include:

- Asking questions about the document
- Extracting relevant information
- Explaining document content
- Analysing the information presented in the document
- Summarizing the document content

2. PPT Generation

2.1. Purpose and Use

The **PPT Generation** service enables users to create presentation content based on a specified topic, requirement, or set of instructions. It can generate structured presentation material according to the information provided by the user.

The service can be used to generate:

- Presentation titles
- Slide-wise content
- Key points
- Examples and supporting information
- Conclusions
- Topic-specific presentation structures

2.2. When to Select PPT Generation

Select **PPT Generation** when the primary requirement is to create presentation content.

Examples:

> "Create a presentation on Artificial Intelligence."
> "Prepare a five-slide presentation on Cybersecurity."
> "Create a presentation on Cloud Computing with introduction, architecture, advantages and conclusion."
> "Prepare presentation content for a technical seminar."

2.3. Creating a Presentation Using PPT Generation

Procedure

1. Open the **AI Service Selection** menu.
2. Select **PPT Generation**.
3. Enter the topic or presentation requirement.
4. Specify the required number of slides, if applicable.

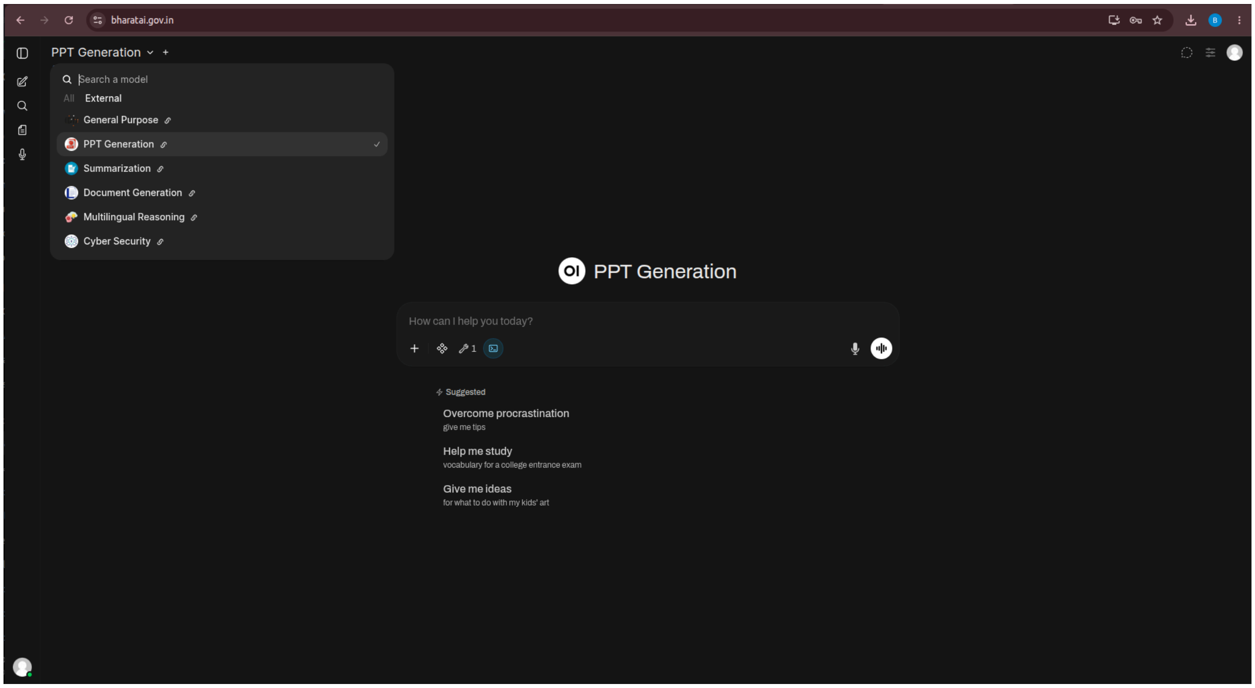


Figure: PPT Generation Service Selection

2.4. Input Types

A. Text Input

Users can provide a **text prompt** describing the presentation requirements. The prompt should clearly specify the desired topic and presentation details to help generate relevant and well-structured content.

A text prompt may include:

- Presentation topic
- Number of slides
- Target audience
- Key sections or topics
- Examples or supporting information
- Presentation objective
- Conclusion or key takeaways

Example:

> “Create a 5-slide presentation on "Artificial Intelligence in Cybersecurity" with a title slide, key points, examples, and conclusion.”

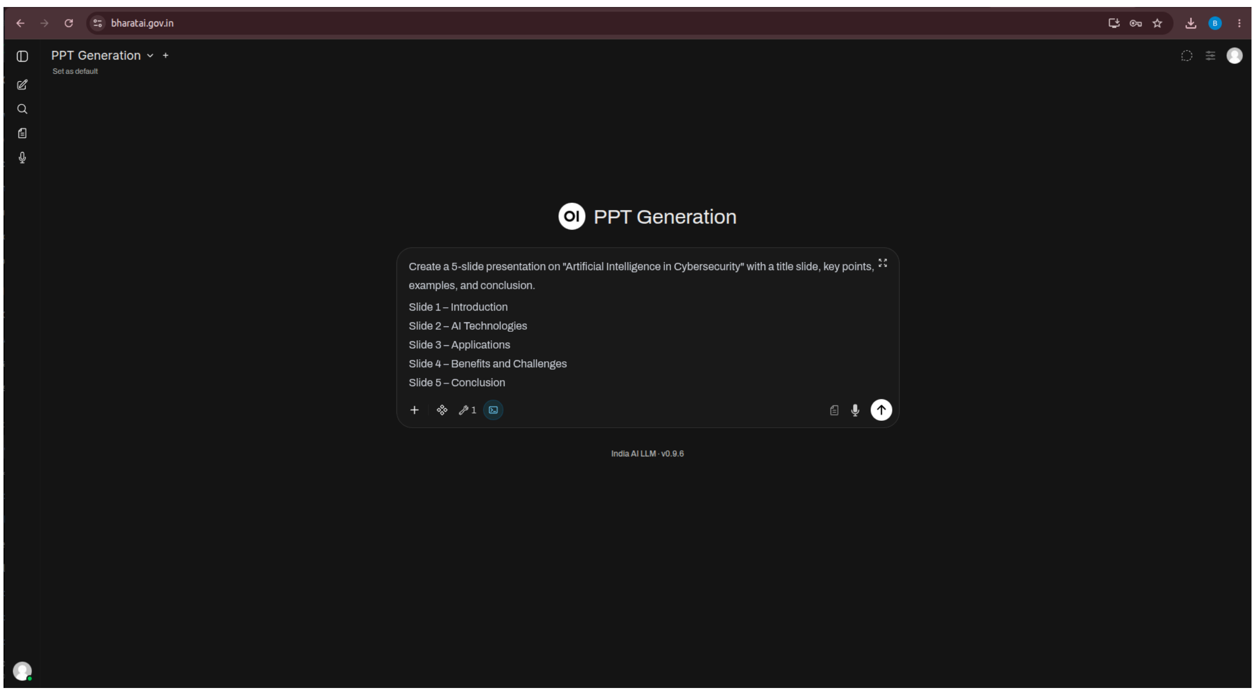
This is the exact type of presentation-generation scenario tested in the report.

B. Presentation Outline

Users can provide an outline specifying the sections or slide structure.

Example:

Slide 1 – Introduction
Slide 2 – AI Technologies
Slide 3 – Applications
Slide 4 – Benefits and Challenges
Slide 5 – Conclusion



3. Summarization

3.1 Purpose and Use

The **Summarization** service enables users to condense lengthy textual or document-based information into a concise representation while retaining the key points and relevant information.

It can be used for:

- Summarizing technical articles
- Summarizing reports and documents
- Extracting key points and findings
- Preparing brief summaries
- Creating executive summaries
- Condensing lengthy information into concise content
- Obtaining quick overviews of detailed information

Users can provide the required content and specify the desired summary format or level of detail based on their requirements.

3.2. When to Select Summarization

Select **Summarization** when the main requirement is to reduce lengthy content into a concise form.

Examples:

> "Summarize this technical article in five points."

3.3. Summarizing Content

Procedure

1. Open the **AI Service Selection** menu.
2. Select **Summarization**.
3. Provide the content to be summarized.
4. Review the generated summary.

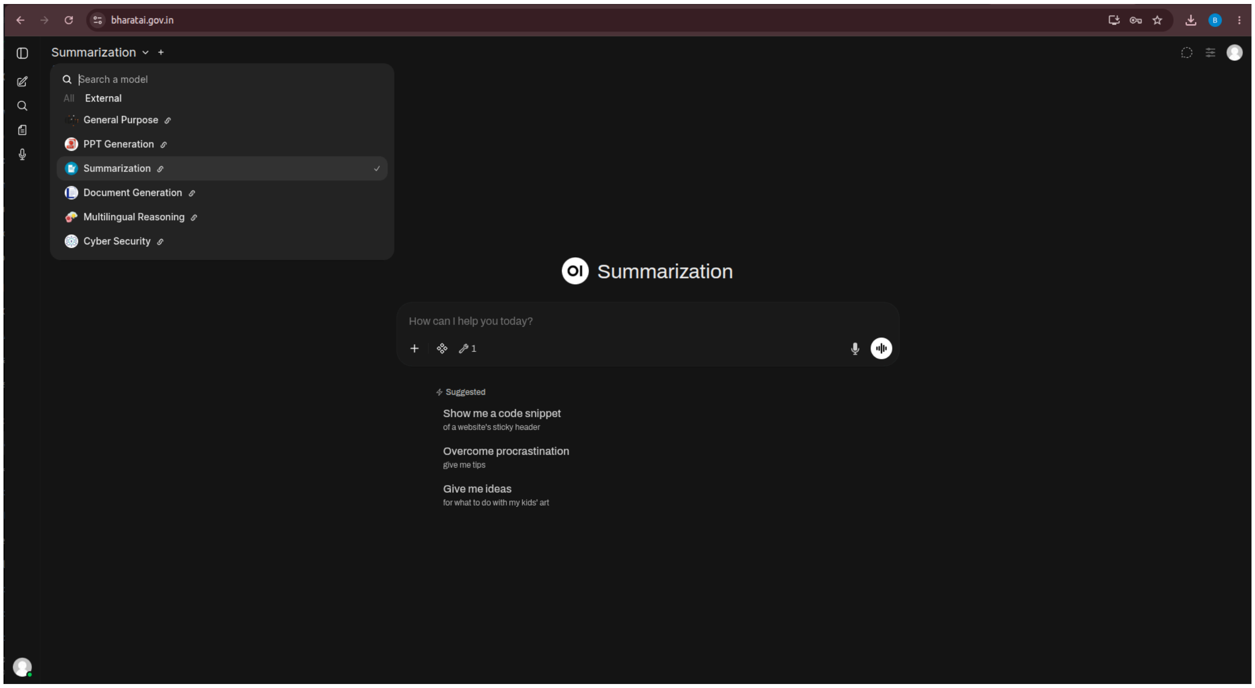


Figure: Summarization Service Selection

3.4. Input Types

A. Text Input

Users can enter or paste **textual content** directly into the input field for summarization. The service processes the provided content and generates a concise summary highlighting the key information.

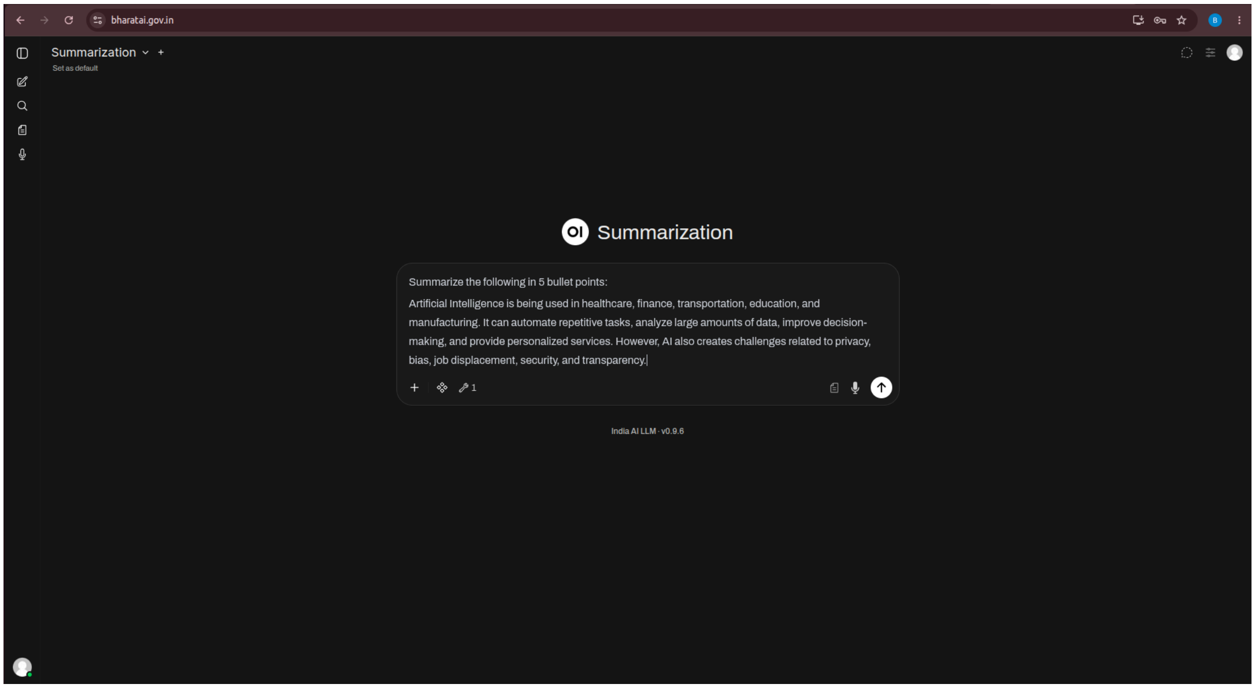
Common text inputs include:

- Articles
- Reports
- Meeting notes
- Technical content
- Policy information
- General textual information

Example:

"Summarize the following in 5 bullet points:
Artificial Intelligence is being used in healthcare, finance, transportation, education, and manufacturing. It can automate repetitive tasks, analyze large amounts of data, improve

decision-making, and provide personalized services. However, AI also creates challenges related to privacy, bias, job displacement, security, and transparency."



B. Document Input

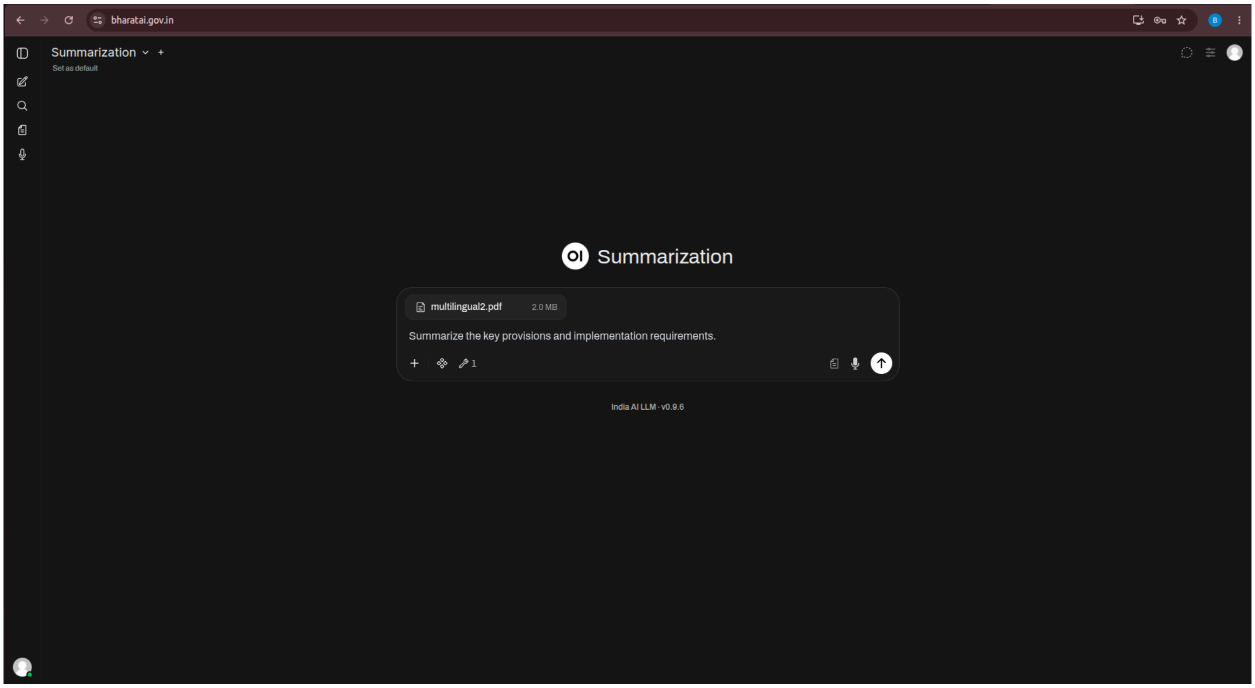
Users can upload **supported documents** for summarization. The uploaded documents are processed to identify and condense the key information into a concise summary.

The document inputs include:

- PDF reports
- Technical documents
- Government orders
- Research documents
- Other supported textual documents

Example:

> Upload a government order and ask: "Summarize the key provisions and implementation requirements."



C. Meeting Notes

Users can provide **meeting notes or discussion content** as text for summarization. The service can identify and present important information from the provided content in a concise and structured manner.

The summary may include:

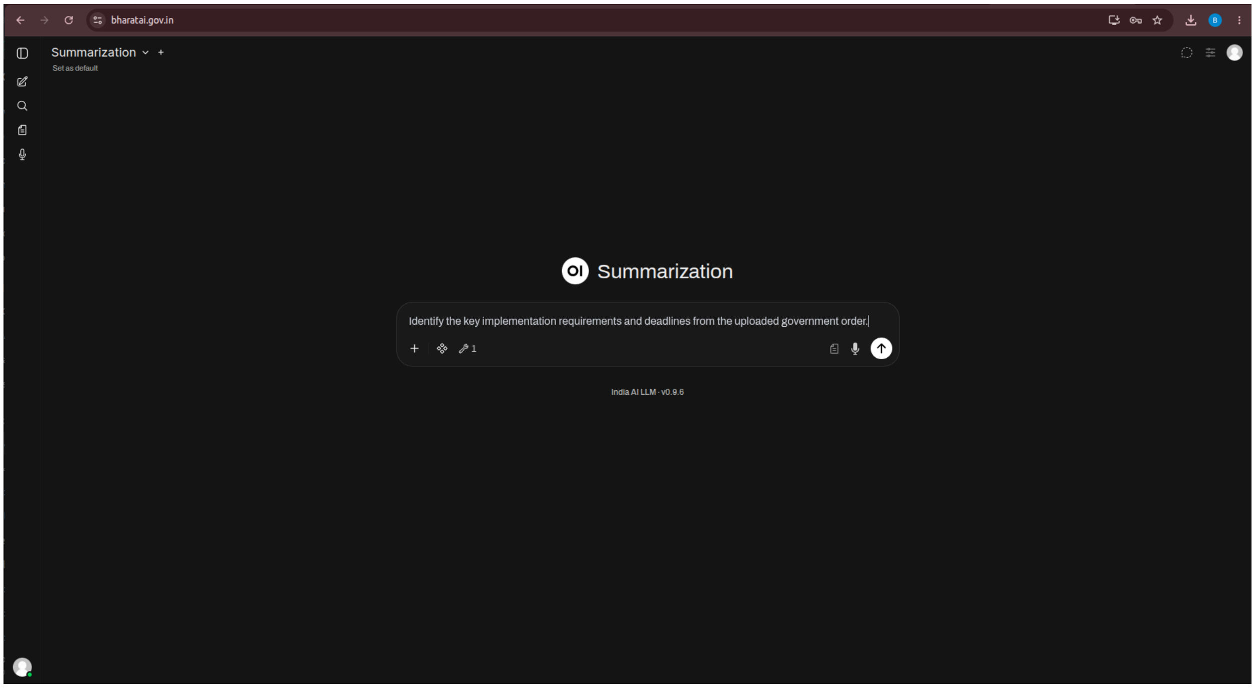
- Key decisions
- Action items
- Responsible persons
- Deadlines and timelines
- Important discussion points

D. Government Orders and Official Documents

Users can provide **government orders and other official documents** for summarization. The service can process the provided content and generate a concise summary highlighting the key provisions, requirements, decisions, and other relevant information.

Example:

> "Identify the key implementation requirements and deadlines from the uploaded government order."



4. Document Generation

4.1 Purpose and Use

The **Document Generation** service enables users to create structured documents based on a specified topic, instruction, outline, or set of requirements.

It can be used to prepare:

- Technical reports
- Project documents
- Structured reports
- Explanatory documents
- Draft documents
- Official document drafts
- Other structured textual documents

Users can specify the required sections, content, structure, and level of detail to generate a document according to their requirements.

4.2. When to Select Document Generation

Select **Document Generation** when the requirement is to create a structured document rather than a presentation or short response.

Examples:

"Create a technical report on Cloud Computing Architecture."
"Prepare a project report with introduction, methodology, results and conclusion."
"Generate a structured document on Artificial Intelligence."
"Prepare a draft technical document."

4.3. Creating a Document

Procedure

1. Open the **AI Service Selection** menu.
2. Select **Document Generation**.
3. Enter the document topic.
4. Specify the required sections.
5. Submit the request.

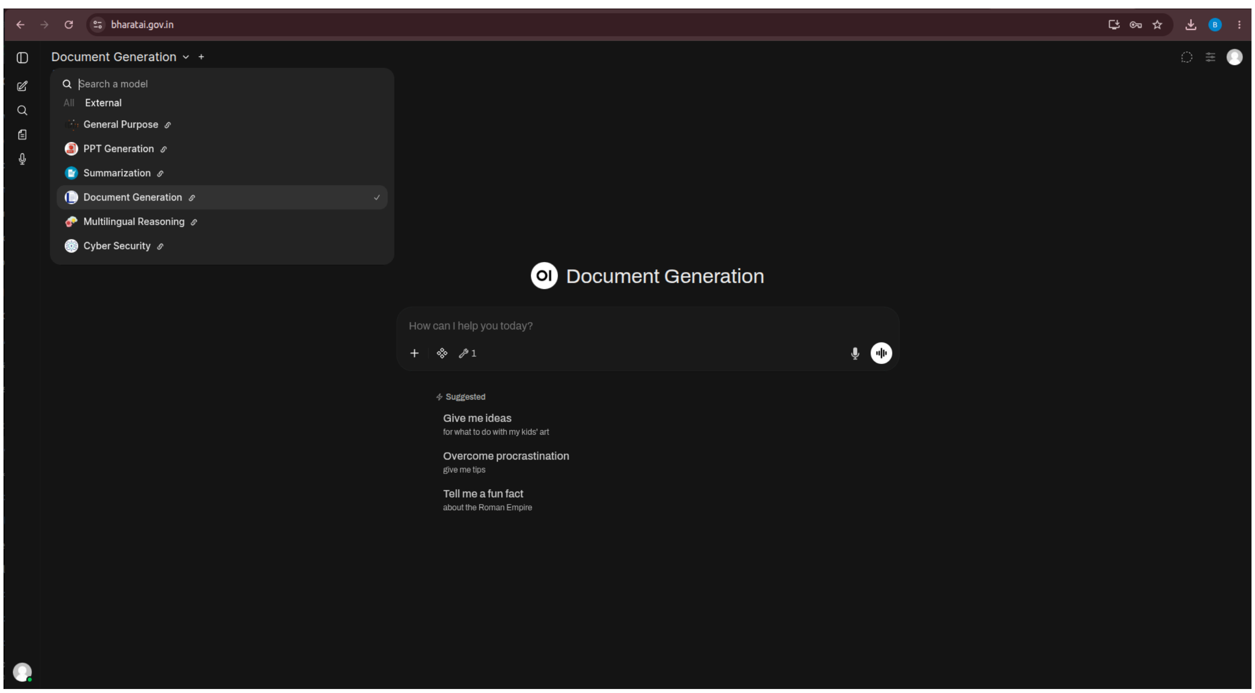


Figure: Document Generation Service Selection

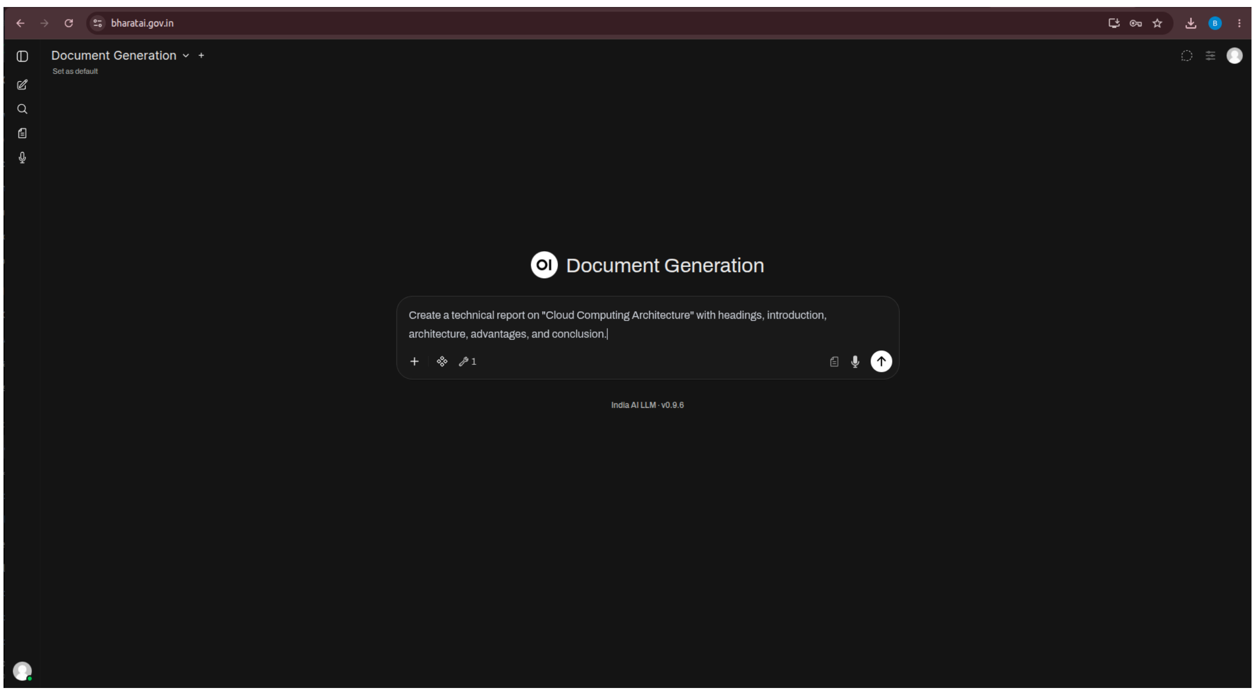
4.4. Input Types

A. Text Input

Users can provide a **text prompt** describing the document they want to generate. The prompt can specify the document topic, purpose, required sections, content, structure, and level of detail to guide the generation of the document.

Example:

> “Create a technical report on "Cloud Computing Architecture" with headings, introduction, architecture, advantages, and conclusion.”



B. Structured Requirements

Users can provide **structured requirements** to define the content and organization of the document. These requirements may include:

- Document title
- Introduction
- Required sections
- Key points
- Technical details
- Conclusion
- Intended audience

- Required level of detail

C. Reference Text

Users can provide **reference information** to guide the content generation process. The provided information can be used as the basis for preparing the requested document.

D. Document Input

Users can upload **supported documents** as reference material for generating the requested content. The uploaded documents can be analysed and used to incorporate relevant information into the generated document.

5. Multilingual Reasoning

5.1 Purpose and Use

The **Multilingual Reasoning** service enables users to interact with the AI system using multiple languages. It supports the understanding, processing, and generation of responses based on multilingual input.

It can be used for:

- Asking questions in different languages
- Receiving responses in the required language
- Analysing multilingual text
- Reasoning over multilingual information
- Working with English and Indian-language content
- Processing mixed-language queries
- Generating content based on multilingual inputs

The service is particularly useful for users who require AI-assisted interaction across different languages and for tasks involving multilingual content.

5.2. When to Select Multilingual Reasoning

Select **Multilingual Reasoning** when the task requires understanding or generating content in more than one language.

Examples:

"What is Machine Learning?"

"Explain Machine Learning in Tamil."
"Explain Machine Learning in Hindi."
"Explain Machine Learning using a mixture of Tamil and English."

5.3. Using Multilingual Reasoning

Procedure

1. Open the **AI Service Selection** menu.
2. Select **Multilingual Reasoning**.
3. Enter the question or instruction.
4. Enter the content in the required language or languages.
5. Specify the preferred response language, where applicable.
6. Submit the request.

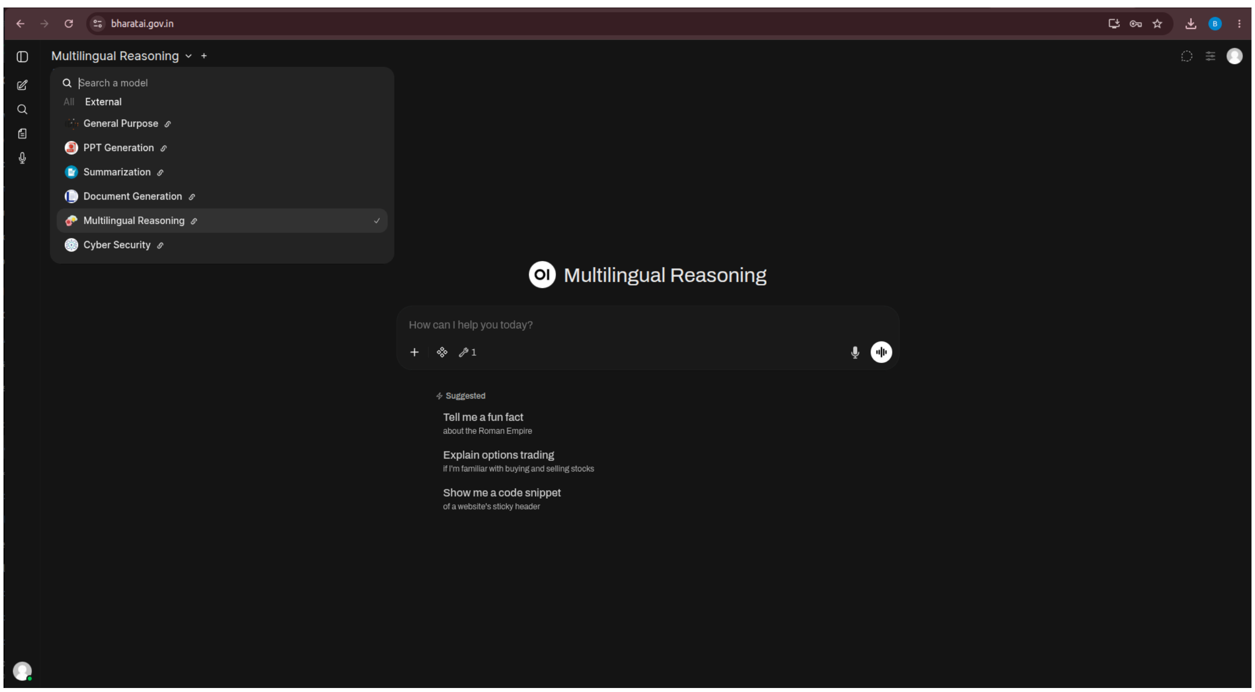


Figure: Multilingual Reasoning Service Selection

5.4. Input Types

A. English Text

Users can enter **questions, instructions, and other textual content in English** for processing and reasoning.

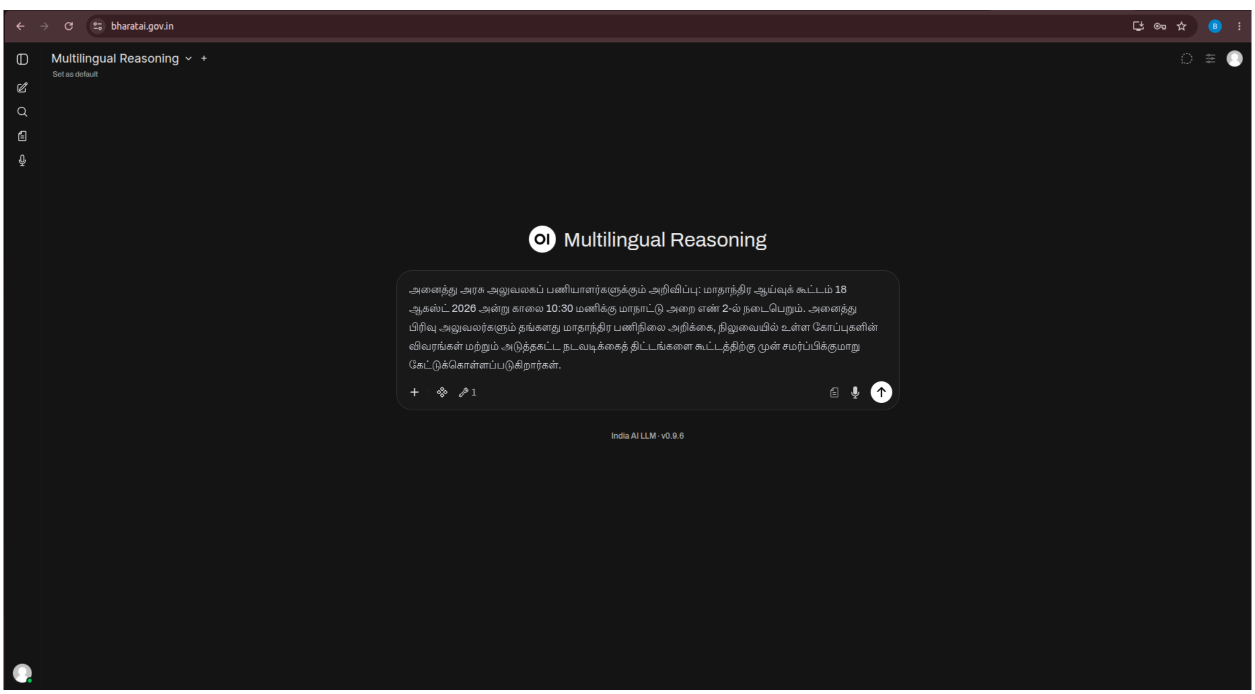
B. Indian-Language Text

Users can enter **questions and instructions in supported Indian languages** for multilingual processing and response generation. The service supports interaction in languages such as **Tamil and Hindi**, as demonstrated through the available platform capabilities.

Example:

“அனைத்து அரசு அலுவலகப் பணியாளர்களுக்கும் அறிவிப்பு: மாதாந்திர ஆய்வுக் கூட்டம் 18 ஆகஸ்ட் 2026 அன்று காலை 10:30 மணிக்கு மாநாட்டு அறை எண் 2-ல் நடைபெறும். அனைத்து பிரிவு அலுவலர்களும் தங்களது மாதாந்திர பணிநிலை அறிக்கை, நிலுவையில் உள்ள கோப்புகளின் விவரங்கள் மற்றும் அடுத்தகட்ட நடவடிக்கைத் திட்டங்களை கூட்டத்திற்கு முன் சமர்ப்பிக்குமாறு கேட்டுக்கொள்ளப்படுகிறார்கள்.”

“All government office employees are hereby informed that the monthly review meeting will be held on 18 August 2026 at 10:30 AM in Conference Room 2. All section officers are requested to submit their monthly work status report, details of pending files, and proposed action plans before the meeting. Employees are requested to be present on time and carry all relevant documents.”

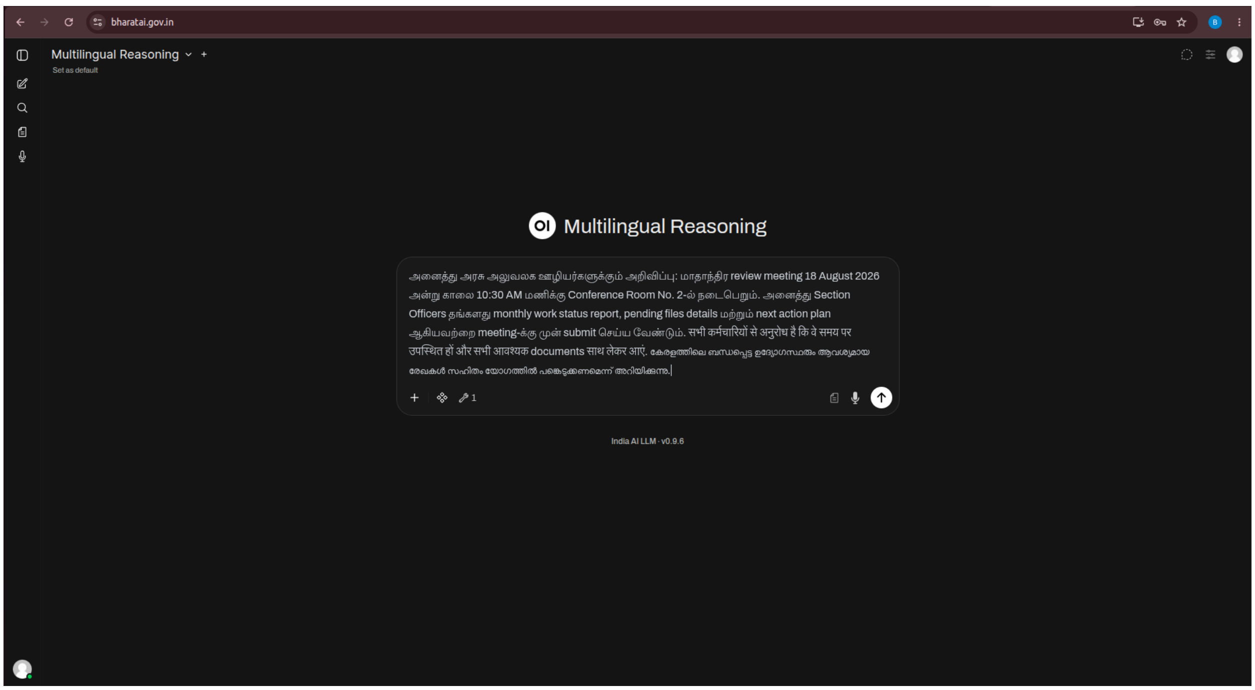


C. Mixed-Language Text

Users can provide a query containing more than one language.

Example:

> "அனைத்து அரசு அலுவலக ஊழியர்களுக்கும் அறிவிப்பு: மாதாந்திர review meeting 18 August 2026 அன்று காலை 10:30 AM மணிக்கு Conference Room No. 2-ல் நடைபெறும். அனைத்து Section Officers தங்களது monthly work status report, pending files details மற்றும் next action plan ஆகியவற்றை meeting-க்கு முன் submit செய்ய வேண்டும். सभी कर्मचारियों से अनुरोध है कि वे समय पर उपस्थित हों और सभी आवश्यक documents साथ लेकर आएं. കേരളത്തിലെ ബന്ധപ്പെട്ട ഉദ്യോഗസ്ഥരും ആവശ്യമായ രേഖകൾ സഹിതം യോഗത്തിൽ പങ്കെടുക്കണമെന്ന് അറിയിക്കുന്നു."



D. Multilingual Document Input

Users can upload **supported documents containing content in multiple languages** for analysis and processing. The service can process the multilingual content and generate relevant responses based on the information provided.

E. Voice Input

The **Voice Input** feature allows users to provide spoken queries using **both male and female voices**. The spoken input is processed to understand the user's request and generate an appropriate response.

6. Cyber Security

6.1 Purpose and Use

The **Cyber Security** service provides AI-assisted capabilities for cybersecurity-related queries and tasks. It enables users to obtain information, explanations, and assistance related to cybersecurity concepts and technical content.

It can be used for:

- Understanding cybersecurity concepts
- Explaining security terminology
- Analysing security-related information
- Understanding security configurations
- Reviewing security-related technical content
- Preparing cybersecurity documentation
- Learning and exploring cybersecurity topics

6.2. When to Select Cyber Security

Select **Cyber Security** when the primary subject of the request is cybersecurity or information security.

Examples:

> "Explain the difference between authentication and authorization."
> "Explain the concept of SQL injection."
> "What are the common security controls for a web application?"
> "Explain the purpose of a firewall."
> "Describe common security considerations for REST APIs."

6.3. Using the Cyber Security Service

Procedure

1. Open the **AI Service Selection** menu.
2. Select **Cyber Security**.
3. Enter the cybersecurity-related question or requirement.
4. Submit the request.

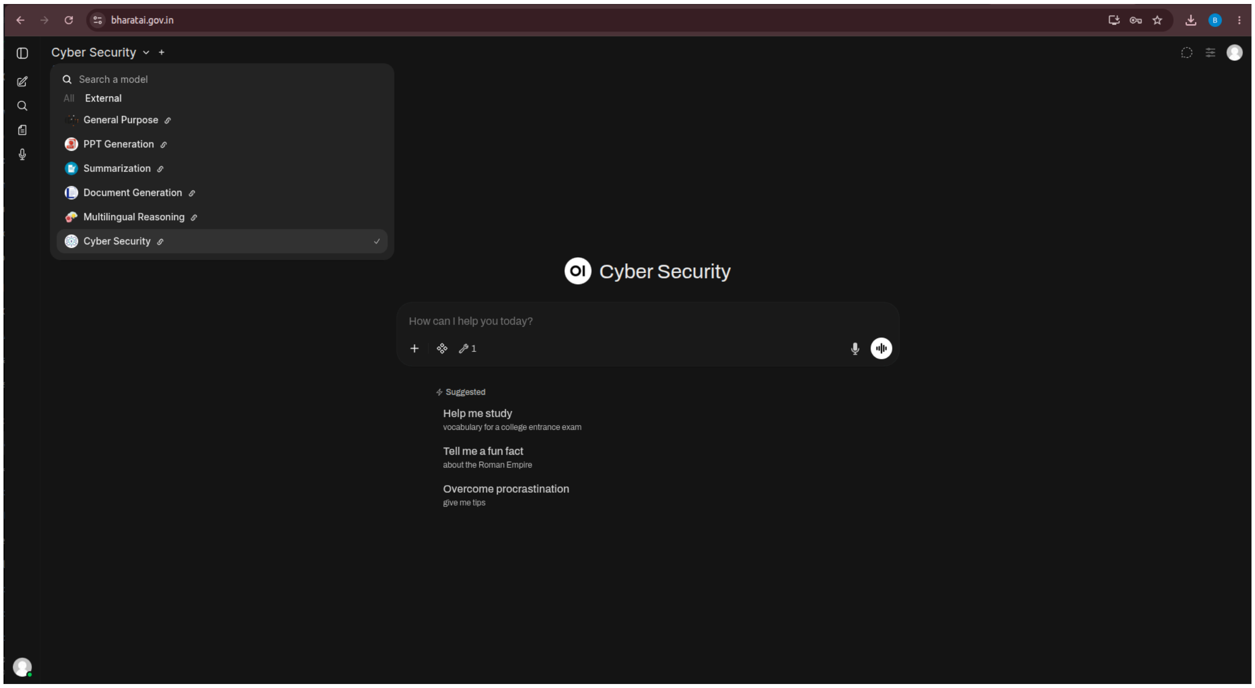


Figure 10.6: Cyber Security Service Selection

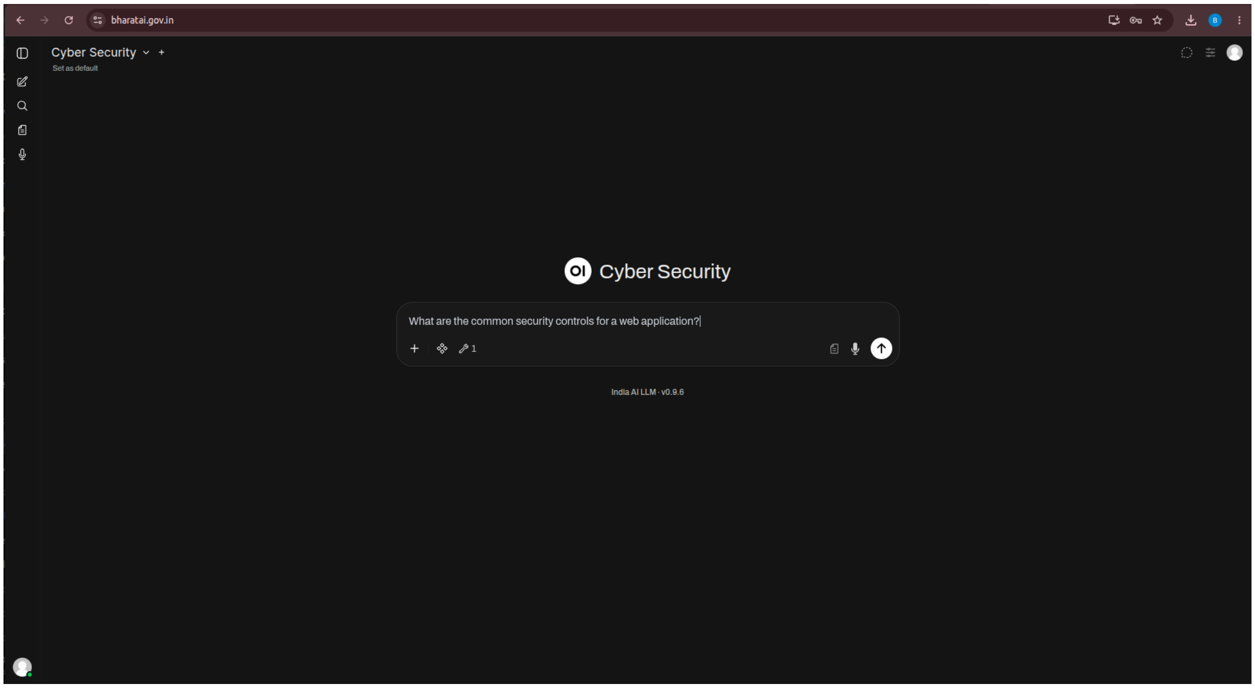
6.4. Input Types

A. Text Input

Users can enter **cybersecurity-related questions, information, and instructions** as text for analysis and response generation.

Typical text inputs include:

- Cybersecurity questions
- Security scenarios
- Technical descriptions
- Configuration-related queries
- Security concepts and terminology
- Error messages and security-related issues



B. Code and Configuration Input

Users can provide **code snippets and configuration details** for cybersecurity-related analysis, review, and explanation.

C. Security Document Input

Users can upload **supported security-related documents** for analysis and extraction of relevant security information.

D. Log and Structured Security Data Input

Users can provide **security logs and structured security information** for analysis and identification of relevant security-related information.

4. System Requirements

The Bharat AI Platform is accessible through a **modern web browser with an active internet connection**. Users should use an updated version of a supported web browser to ensure proper functioning of the platform interface and its AI services.

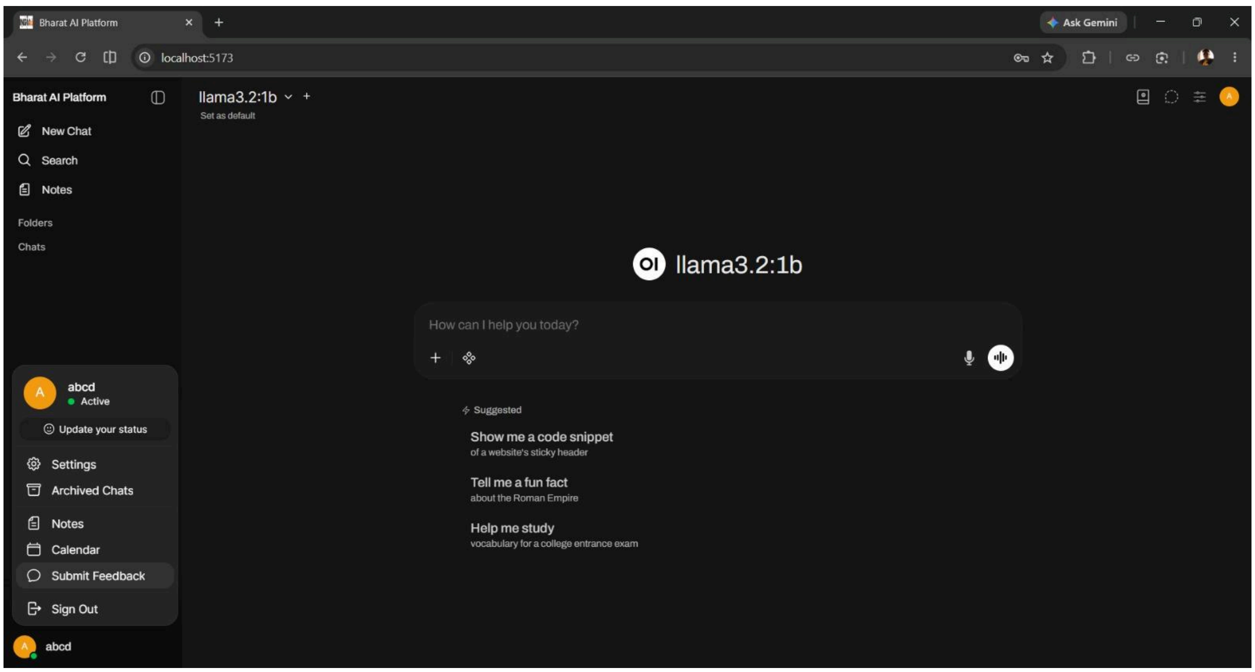
System requirements may vary depending on the selected service and the type of input being processed. Users should ensure that their system meets the required browser and connectivity requirements for accessing and using the platform effectively.

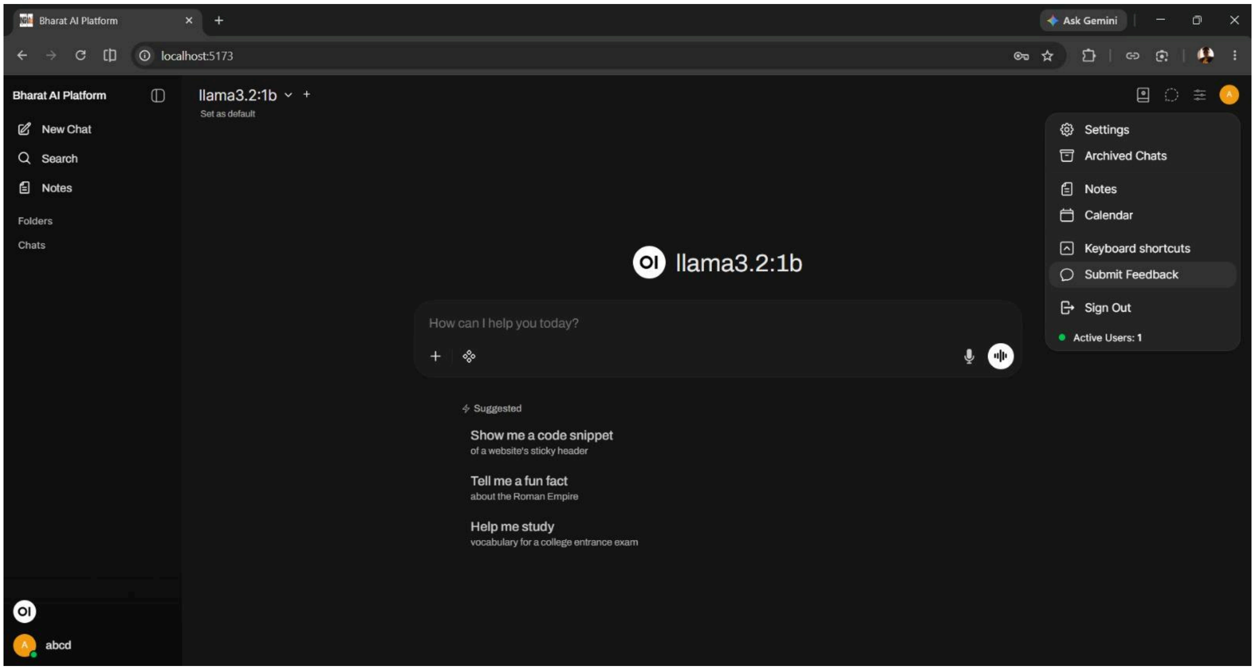
5. Feedback and Suggestion

The Feedback and Suggestions feature enables users to provide feedback on their experience with the Bharat AI Platform. Users can share their comments, suggestions, or observations regarding the platform, AI services, usability, and overall user experience.

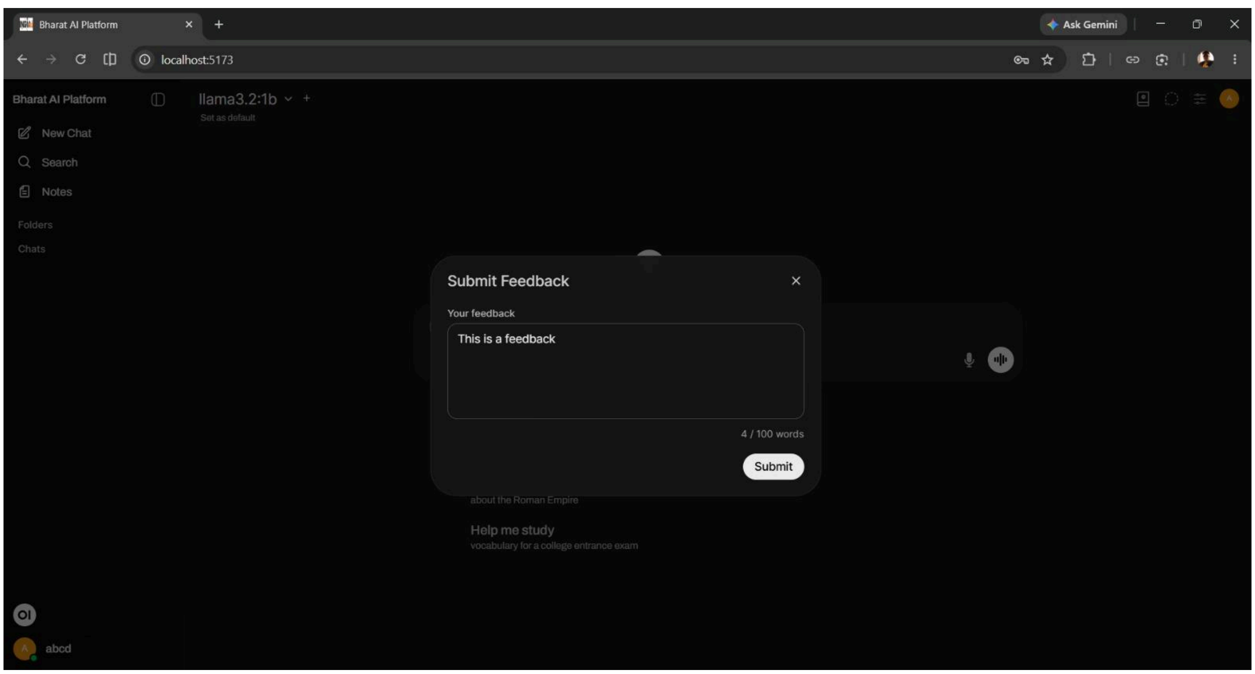
Users are encouraged to provide clear and relevant feedback to help identify areas for improvement and support the continuous enhancement of the platform and its services.

Where applicable, users should provide sufficient details about their feedback, including the relevant AI service, description of the experience, and suggestions for improvement.

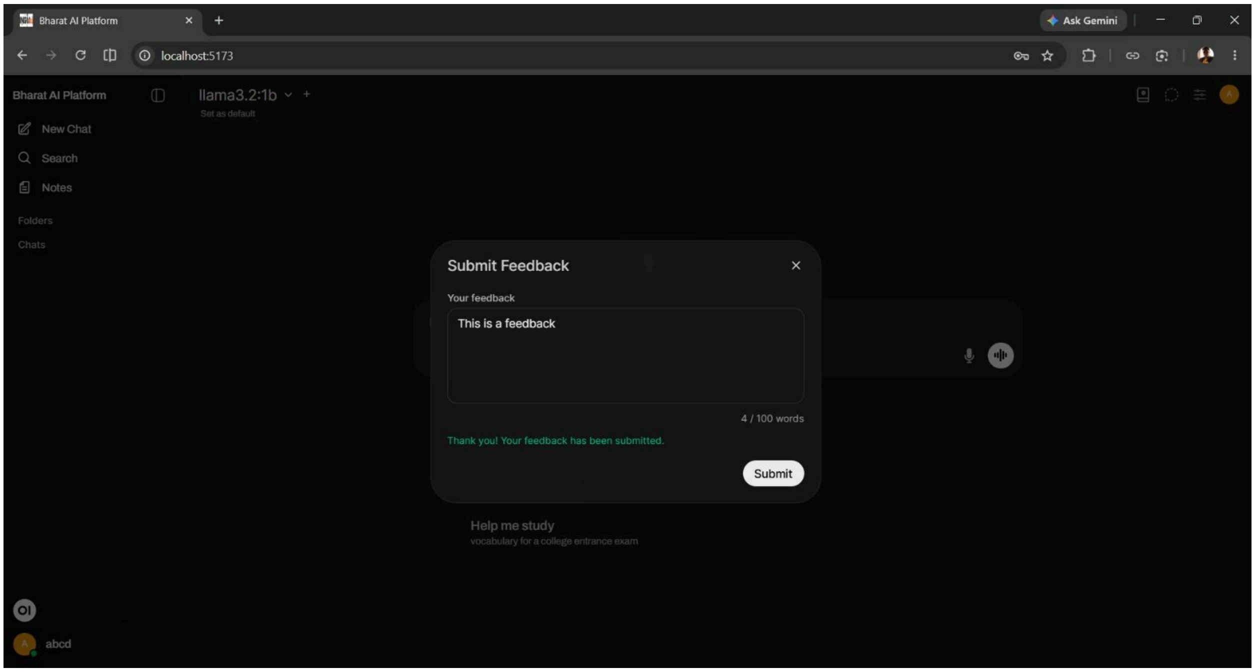




Bharat AI Platform
localhost:5173
Bharat AI Platform
New Chat
Search
Notes
Folders
Chats
llama3.2:1b
Set as default
Settings
Archived Chats
Notes
Calendar
Keyboard shortcuts
Submit Feedback
Sign Out
Active Users: 1
llama3.2:1b
How can I help you today?
Suggested
Show me a code snippet
of a website's sticky header
Tell me a fun fact
about the Roman Empire
Help me study
vocabulary for a college entrance exam
abcd



Bharat AI Platform
localhost:5173
Submit Feedback
Your feedback
This is a feedback
4 / 100 words
Submit



6. Glossary

Term	Definition
AI	Artificial Intelligence, a technology that enables computer systems to perform tasks that normally require human intelligence.
AI Service	A platform-provided capability designed for a specific AI-related task.
Prompt	A question, instruction, or input provided to an AI service.
LLM	Large Language Model, an AI model designed to understand and generate human language.
Multilingual Reasoning	AI-assisted processing and reasoning involving multiple languages.
Voice Input	A feature that allows users to provide spoken input to the platform.
Session	An ongoing interaction between the user and an AI service.
Dashboard	The primary interface through which users access platform services and functions.